

System Requirements

WEB-BASED BUSINESS MANAGEMENT SYSTEM

2. System Requirements

2.1 Functional Requirements

The system shall provide the following functionalities:

User Management

FR1	The system shall allow business owners to create user accounts.
FR2	The system shall authenticate users using email and password.
FR3	The system shall support role-based access control (Owner and Cashier).
FR4	The system shall allow authorized users to update their profiles and passwords.
FR5	The system shall allow owners to activate or deactivate user accounts.

Business Management

FR6	The system shall allow the registration and management of business information.
FR7	The system shall store business details such as name, address, contact information, logo, and tax PIN.
FR8	The system shall allow customization of receipt information.

Customer Management

FR9	The system shall allow users to register customers.
FR10	The system shall store customer details including name, phone number, email, and address.
FR11	The system shall allow searching customers by name or phone number.
FR12	The system shall support sales without requiring customer registration.

Supplier Management

FR13 The system shall allow users to register suppliers.

FR14 The system shall store supplier contact information.

FR15 The system shall allow updating supplier details.

Category Management

FR16 The system shall allow creation of product categories.

FR17 The system shall allow editing and deleting product categories.

Product Management

FR18 The system shall allow users to add products.

FR19 The system shall allow editing product information.

FR20 The system shall allow product deactivation instead of permanent deletion.

FR21 The system shall maintain stock levels for every product.

FR22 The system shall support barcode and SKU management.

Inventory Management

FR23 The system shall automatically update stock quantities after sales.

FR24 The system shall increase stock after approved purchase orders are received.

FR25 The system shall record all inventory movements.

FR26 The system shall support stock adjustments.

FR27 The system shall notify users when stock falls below the reorder level.

Purchase Order Management

FR28 The system shall allow users to create purchase orders.

FR29 The system shall allow adding multiple products to a purchase order.

FR30 The system shall track the status of purchase orders.

FR31 The system shall record goods received from suppliers.

Sales Management

FR32	The system shall allow cashiers to process sales.
FR33	The system shall support multiple products in one sale.
FR34	The system shall calculate subtotals, discounts, and total amounts automatically.
FR35	The system shall support sales with or without registered customers.
FR36	The system shall automatically reduce inventory after completed sales.

Payment Management

FR37	The system shall record payments.
FR38	The system shall support cash, M-Pesa, card, and bank transfer payments.
FR39	The system shall record payment references where applicable.

Receipt Management

FR40	The system shall generate unique receipt numbers.
FR41	The system shall generate printable receipts.
FR42	The system shall support digital receipt delivery via email.
FR43	The system shall maintain receipt history.

Reporting

FR44	The system shall generate daily sales reports.
FR45	The system shall generate inventory reports.
FR46	The system shall generate purchase reports.
FR47	The system shall display low-stock reports.
FR48	The system shall allow report filtering by date.

2.2 Non-Functional Requirements

Performance Requirements

NFR1	The system shall process sales transactions within three seconds under normal operating conditions.
NFR2	The system shall support multiple users simultaneously without significant performance degradation.
NFR3	The system shall return search results within two seconds.

Security Requirements

NFR4	User passwords shall be securely hashed before storage.
NFR5	Only authenticated users shall access the system.
NFR6	Access shall be controlled according to user roles.
NFR7	Sensitive data shall be transmitted securely using HTTPS in production.

Reliability Requirements

NFR8	The system shall maintain data integrity through database constraints and transactions.
NFR9	The system shall prevent duplicate records where uniqueness is required.
NFR10	The system shall support database backup and restoration.

Usability Requirements

NFR11	The user interface shall be simple and easy to navigate.
NFR12	Users shall be able to learn the basic functions with minimal training.
NFR13	Error messages shall clearly describe problems and suggest corrective actions.

Maintainability Requirements

NFR14	The application shall use a modular architecture.
NFR15	Database scripts shall be organized into separate SQL files.
NFR16	The codebase shall be documented to simplify future maintenance.

Scalability Requirements

NFR17	The system shall support future implementation of multi-branch management.
NFR18	The database shall support future expansion without major redesign.
NFR19	The architecture shall support integration with additional services such as SMS and WhatsApp notifications.

Compatibility Requirements

NFR20	The system shall support modern web browsers such as Chrome, Firefox, Edge, and Safari.
NFR21	The user interface shall be responsive on desktops, tablets, and smartphones.